

NeuroDrift: A Gated, Longitudinally-Aware Pipeline for Motor-Imagery EEG Classification

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Abstract—Motor-imagery brain-computer interfaces (BCIs) suffer from inter-session signal non-stationarity that degrades classifier performance over longitudinal use without any observable model failure. NeuroDrift addresses this with a three-stage pipeline: (1) a frozen logistic-regression *acceptance gate* rejects artifact-contaminated calibration trials before they corrupt covariance estimation (test AUC=0.688, +4.2 pp accuracy lift); (2) an adaptive stopping controller terminates calibration as soon as five orthogonal quality criteria plateau (5-fold CV accuracy=0.832 ± 0.072); and (3) a Ridge regression module tracks session-level drift via a scalar recovery factor $r \in [0, 1]$, achieving LOO $R^2=0.997$ (RMSE=0.022). LOSO evaluation across nine subjects yields $\mu=0.728$ ($\sigma=0.174$), exposing substantial inter-subject heterogeneity. A key scientific contribution is that LDA classifier accuracy is a *poor* proxy for signal quality (Cohen’s $d=1.011$, $p=0.238$; $\Delta R^2 \approx 0$ in drift regression); raw physiological signal features anchored to absolute plausibility thresholds are strictly more reliable drift indicators.

I. INTRODUCTION

The domain of this project is **Health Care**. Motor-imagery BCI systems translate imagined limb movement—detected as event-related desynchronization (ERD) in mu (8–12 Hz) and beta (13–30 Hz) EEG bands over sensorimotor cortex—into control signals that bypass damaged motor pathways. Despite laboratory success, clinical deployment is impeded by systematic inter-session signal non-stationarity arising from electrode impedance variation, subject fatigue, and neurological variability [1].

The dominant calibration paradigm—record a session-specific block, fit a CSP spatial filter and LDA classifier, and evaluate on subsequent trials—makes two critical assumptions that fail in longitudinal use: (i) calibration trials are representative of the session distribution, and (ii) that distribution is stationary across sessions. Neither assumption holds in practice.

NeuroDrift directly addresses both assumptions through three coordinated pipeline stages. The acceptance gate (Stage 1) rejects artifact-contaminated calibration trials before they corrupt covariance estimation. The adaptive calibration controller (Stage 2) stops collecting calibration data as soon as a statistically reliable plateau is detected. The longitudinal module (Stage 3) tracks session-level feature drift via regression and clustering, enabling early detection of clinical regression without requiring continuously labelled data.

II. PROBLEM STATEMENT

Brain-computer interface classifiers exhibit well-documented inter-session performance degradation without observable model failure—a failure mode distinct from standard overfitting. The primary cause is EEG non-stationarity: electrode impedance drift, subject fatigue, and neurological variability systematically shift sensorimotor mu and beta rhythms in frequency and scalp topography across sessions, the clinically relevant timescale for rehabilitation monitoring.

Standard calibration protocols rest on two assumptions that fail in practice. First, calibration trials are assumed artifact-free; in reality electrode pops, ocular artifacts, and muscle contamination corrupt the covariance estimates that CSP spatial filtering relies upon. Second, the session distribution is assumed stationary; longitudinal drift violates this across days or weeks. Existing mitigations—transfer learning, Riemannian geometry adaptation, and ensemble classifiers—address cross-session variability but neither explicitly model signal quality degradation nor provide interpretable drift metrics for clinical monitoring. NeuroDrift addresses both assumptions directly with a three-stage pipeline: (1) a frozen acceptance gate rejects artifact-contaminated trials before classifier training; (2) a rule-based stopping controller adapts the calibration budget to session-specific signal quality; and (3) a Ridge regression module tracks longitudinal drift over session-level feature representations, enabling quantitative monitoring without continuous expert labelling.

NeuroDrift introduces a scalar recovery factor $r \in [0, 1]$ encoding session-level signal quality relative to a pristine baseline session ($s = 0$), and learns a mapping from a 16-dimensional session feature vector ϕ_s to $\hat{r}_s$ via Ridge regression.

III. METHODOLOGY

A. Dataset

All classification experiments use the BNCI2014_001 dataset accessed via MOABB [2], comprising 22-channel EEG recorded at 250 Hz during a left/right hand motor-imagery paradigm. Epoch windows span -2 s to $+4$ s relative to cue onset; the task window is $[0, +4]$ s (samples 500–1750) and the baseline window is $[-2, 0]$ s (samples 0–499). Subjects 2–9 (1,152 trials; 576 per class) constitute the global training pool; Subject 1 (288 trials; 144 calibration / 144 evaluation) is

the held-out evaluation subject for Stage 2. LOSO evaluation is additionally performed across all nine subjects.

B. Stage 1 — Global Acceptance Gate

For each epoch $\mathbf{X}_t$, an 8-dimensional raw quality feature vector $\mathbf{q}_t \in \mathbb{R}^8$ is extracted (Table I). Features combine amplitude statistics, kurtosis-based spike detection, and physiologically grounded ERD measures derived from Welch power spectral density estimates in the mu band at electrodes C3 and C4.

A Logistic Regression classifier with balanced class weighting is trained on 3,352 trials (942 genuine, 2,410 synthetic negatives generated by a CorruptionEngine). The model outputs $P(\text{accept}|\mathbf{q}_t)$; the operating threshold is the 30th percentile of the calibration-set $P(\text{accept})$ distribution, admitting the top 70% of trials by acceptance score. The gate is trained once on the global pool and frozen at deployment.

C. Stage 2 — Adaptive CSP+LDA Calibration

Gate-accepted trials enter an incremental calibration loop. At each batch boundary $n \in \{48, 60, 72, 84, \dots\}$, CSP with $K = 6$ components is fit on the task window, yielding filter $\mathbf{W} \in \mathbb{R}^{6 \times 22}$. Log-variance features are extracted as:

$$f_k = \log \text{Var}(\mathbf{W}_k \cdot \mathbf{X}), \quad k = 1, \dots, K. \quad (1)$$

An LDA classifier is evaluated via 5-fold cross-validation. Early stopping is triggered when five orthogonal criteria are all satisfied for two consecutive batches (patience=2): CV accuracy ≥ 0.72 ; mean confidence ≥ 0.62 ; confidence std ≤ 0.14 ; rejection rate ≤ 0.35 ; Frobenius covariance drift ≤ 0.80 . The final deployment model uses CSP with $K = 4$ components.

D. Stage 3 — Longitudinal Drift Tracking

Each session s is summarised by a 16-dimensional feature vector ϕ_s concatenating CSP log-variances, LDA margin statistics, session accuracy, Mahalanobis distance from the session centroid, raw signal quality means, and Frobenius distance from the Session-0 covariance baseline. A Ridge regression model maps $\phi_s \rightarrow r_s$; KMeans ($k = 3$) clusters sessions into impaired / recovering / pristine trajectories. An unsupervised KMeans ($k = 2$) profiler stratifies subjects into healthy vs. patient domains.

IV. ML EXPERIMENTS — RESULTS AND ANALYSIS

A. Stage 1: Acceptance Gate Performance

The gate was trained on 3,352 trials (2,304 synthetic hard negatives; 942 genuine positives; 106 borderline negatives). Test-set AUC-ROC was 0.711 (train AUC 0.672), indicating cross-subject generalisation of learned quality criteria (Fig. 1a). At threshold 0.5, the reject class achieved F1=0.69 (precision 0.84, recall 0.58) and accept class F1=0.52 (precision 0.41, recall 0.72). The confusion matrix yielded 281 true rejects, 201 false accepts, 52 false rejects, and 137 true accepts.

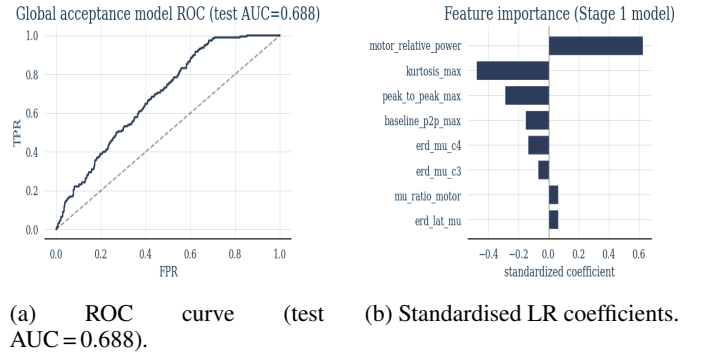


Fig. 1: Stage 1 acceptance gate diagnostics. Green bars push toward acceptance; red bars push toward rejection.

Standardized logistic coefficients (Fig. 1b) show that `motor_relative_power` ($\beta = +0.622$) is the dominant positive predictor, and `kurtosis_max` ($\beta = -0.489$) is the primary artifact rejecter. The negative coefficient on `erd_mu_c4` ($\beta = -0.113$) is counterintuitive and likely reflects spurious apparent desynchronisation in corrupted trials. Corrupt-and-catch validation on 432 trials achieved an overall rejection rate of 38% (163/432); per-family rates ranged from 18% (implausibility) to 47% (disengagement).

B. Stage 2: Per-Subject Calibration (Subject 1)

Of 144 calibration trials submitted to the gate, 101 (70.1%) were accepted. Adaptive stopping ran all four batch steps ($n \in \{48, 60, 72, 84\}$) without triggering, as LDA CV accuracy oscillated between 0.755 and 0.832 without satisfying the patience=2 plateau criterion (Fig. 2). Pool exhaustion at $n = 84$ served as the terminal condition.

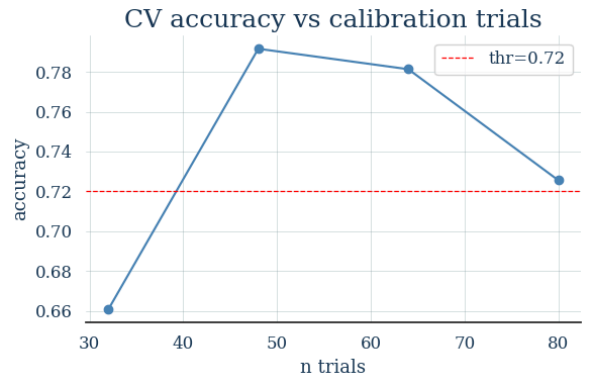


Fig. 2: Adaptive stopping history for Subject 1. Pool was exhausted at $n = 84$ without triggering the patience criterion; both CV accuracy and confidence plateaued just above their respective thresholds (dashed lines).

The final CSP+LDA model (4 components, 84 accepted calibration trials) achieved in-sample accuracy of 0.869 and 5-fold CV accuracy of 0.832 ± 0.072 , a 4-point generalisation gap substantially smaller than typical BCI overfitting patterns. The gated pipeline yielded pseudo-online eval accuracy of

TABLE I: Raw Quality Features (Stage 1 Gate Input), Global Pool Statistics ($N = 1,152$).

Feature	Mean	Std	Physiological Role
peak_to_peak_max	47.34	16.86	Artifact severity
kurtosis_max	1.07	1.20	Impulsive spikes
baseline_p2p_max	46.80	18.18	Pre-trial contamination
erd_mu_c3	-0.082	0.708	Left-motor ERD (C3)
erd_mu_c4	-0.093	0.661	Right-motor ERD (C4)
erd_lat_mu	0.011	0.731	Lateralisation index
mu_ratio_motor	0.754	0.106	Motor mu selectivity
motor_relative_power	1.001	0.098	Motor vs. frontal power

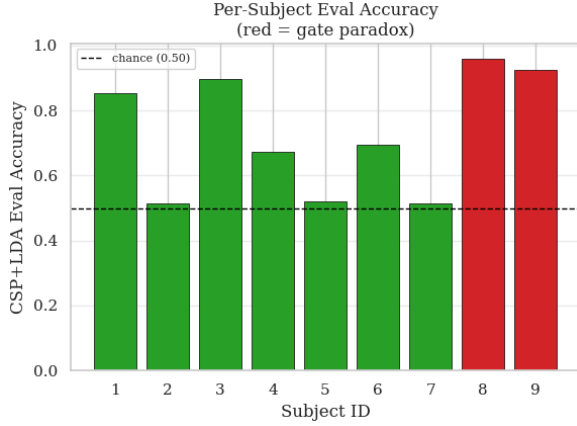


Fig. 3: LOSO evaluation accuracy per subject. Red bars fall below 0.60 (near chance); blue bars are moderate (0.60–0.80); green bars exceed 0.80. The dashed line marks the population mean ($\mu = 0.728$).

0.868 vs. 0.826 for the unfiltered baseline, a +4.2pp lift attributable to excluding 43 artifact-contaminated calibration trials. Accepted eval trials outperformed rejected ones (0.882 vs. 0.800), confirming gate calibration correctness. The of-line/online match test returned PASS.

C. LOSO Inter-Subject Variability

Table II reports LOSO results across all nine subjects (see also Fig. 3). Population statistics are: $\mu = 0.728$, $\sigma = 0.174$, range = [0.514, 0.958], coefficient of variation = 0.239.

The gate paradox (rejected accuracy > accepted accuracy) was observed for Subjects 8 and 9, both of which already achieve near-ceiling accuracy (≥ 0.924) and present low calibration thresholds (0.408 and 0.285, respectively). The non-monotonic relationship between `cal_threshold` and eval accuracy confirms that gate confidence proxies artifact-based signal integrity, not motor imagery discriminability.

D. Stage 3: Longitudinal Regression and Clustering

Ridge LOO regression on the scalar recovery factor r over 18 synthetic sessions achieved $R^2 = 0.997$ and RMSE = 0.022 (Fig. 5). Permutation feature importance identified raw signal quality metrics as dominant predictors: `p2p_max_mean` ($\Delta R^2 = +0.084$), `frob_dist_to_baseline` (+0.054),

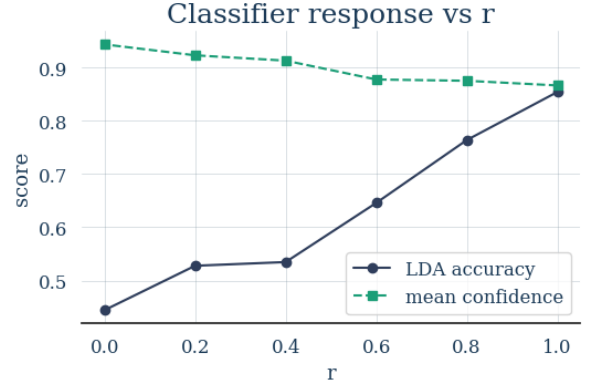


Fig. 4: Stage 3 degradation model: classifier accuracy vs. recovery factor r . LDA confidence remains elevated even at low r , confirming classifier blindness to drift.

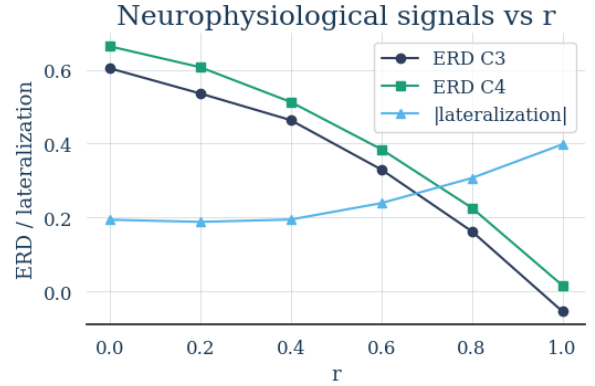


Fig. 5: Stage 3 recovery factor regression: neuro-physiological signals vs. true r . Ridge LOO achieves $R^2 = 0.997$, RMSE = 0.022.

`motor_relative_mean` (+0.035); classifier-derived features (`lda_session_acc`, `lda_conf_mean`) contributed approximately zero. KMeans clustering ($k = 3$) yielded a silhouette score of 0.535, with clusters interpretable as impaired / recovering / near-pristine trajectory groups.

E. Multi-Output Regression and Domain Profiling

A multi-output Ridge model separately regressed four degradation axes. Amplitude (α : LOO $R^2 = 0.997$) and spatial blurring (γ : $R^2 = 0.990$) were recovered with high fidelity;

TABLE II: LOSO Calibration Results Across All Nine Subjects.

Subj.	cal_thr	CV Acc (LDA)	Eval Acc	Acc (Acc.)	Acc (Rej.)	Paradox
1	0.508	0.822 ± 0.023	0.854	0.858	0.833	No
2	0.640	0.464 ± 0.107	0.514	0.527	0.506	No
3	0.438	0.960 ± 0.037	0.896	0.939	0.839	No
4	0.537	0.675 ± 0.100	0.674	0.701	0.617	No
5	0.504	0.505 ± 0.114	0.521	0.521	0.521	No
6	0.599	0.772 ± 0.042	0.694	0.720	0.647	No
7	0.486	0.663 ± 0.061	0.514	0.516	0.510	No
8	0.408	0.980 ± 0.024	0.958	0.957	0.964	Yes
9	0.285	0.772 ± 0.069	0.924	0.915	0.940	Yes

All subjects: $n_{\text{acc,cal}} = 101$ (70.1%), $n_{\text{cal,used}} = 84$ (pool-exhaustion boundary).
 $\mu = 0.728$, $\sigma = 0.174$, range = $[0.514, 0.958]$.

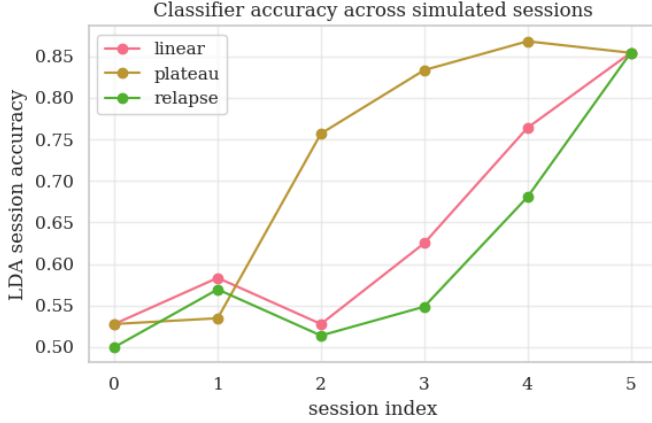


Fig. 6

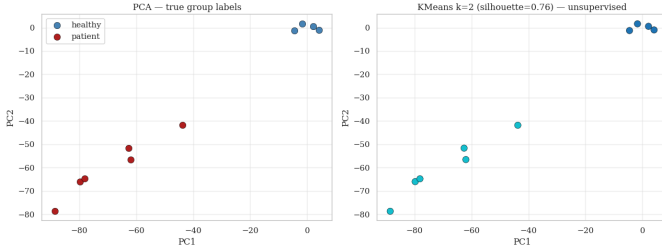


Fig. 7: Healthy vs. patient domain profiling via unsupervised KMeans ($k = 2$, silhouette=0.821). Left: post-hoc ground-truth labels; right: cluster assignments. Perfect cluster-label alignment is achieved without using any group labels during fitting.

noise floor (β : $R^2 = 0.443$) was partially recoverable; timing jitter (δ_{ms} : $R^2 = -0.214$) was not modelled by the current feature set. Cross-prior generalisation yielded $R^2 = 0.601$ vs. 0.999 in-distribution, a gap of 0.398.

Unsupervised KMeans ($k = 2$) domain profiling on 10 synthetic sessions (4 healthy: $r \in [0.85, 1.00]$; 6 patient: $r \in [0.00, 0.25]$) achieved silhouette=0.821 with perfect post-hoc cluster-label alignment (Fig. 7). Top discriminating features by Cohen’s d : motor_relative_mean ($d = +10.911$), frob_dist_to_baseline ($d = -9.953$), erd_mu_c3_mean ($d = -9.925$). lda_session_acc

was not a significant discriminator ($d = 1.011$, $p = 0.238$).

Table III summarises all model performance metrics.

V. SYSTEM ARCHITECTURE

The NeuroDrift codebase is organised as a Python package (`src/neurodrift/`) with strict separation of concerns across submodules: data ingestion, feature extraction, model fitting, serving, orchestration flows, and registry I/O. This separation permits the same logic to be exercised by unit tests, exposed via FastAPI endpoints, and invoked as Prefect tasks without duplication.

Fig. 8 presents the end-to-end NeuroDrift pipeline architecture, encompassing data ingestion, quality gating, per-subject calibration, longitudinal drift tracking, and the DevOps serving infrastructure.

Trained models are persisted as versioned joblib artifacts under `artifacts/registry/`, indexed by a manifest JSON. The FastAPI serving layer exposes six endpoints (Table IV), with Pydantic-validated request/response schemas and auto-generated OpenAPI documentation. A shared Docker volume bridges training (the Prefect worker) and serving (the API container), eliminating a discrete model deployment step.

VI. CONTAINERISATION WORKFLOW

The Docker Compose stack coordinates three services: (i) `api`—a `python:3.11-slim` image running Uvicorn on port 8000 with a 30-second HEALTHCHECK against `/health`, a non-root `neurodrift` user, and a mounted `artifacts/` volume; (ii) `prefect-server`—the `prefect/prefect:2.18` image exposing the Prefect UI on port 4200; and (iii) `prefect-worker`—a custom image that executes training flows and writes versioned artifacts to the shared registry volume. A second shared volume, `moabb_data/`, caches raw data to avoid redundant downloads.

All Prefect tasks carry `retries=3`, `retry_delay_seconds=30` decorators, providing automatic recovery from transient MOABB API failures and network instability. The email notification subsystem is wrapped in a non-fatal `try/except` envelope so that absent SMTP credentials do not abort the flow.

TABLE III: Comparative Model Performance Summary.

Model	Task	Best Metric
LR Gate	Trial accept/reject	AUC=0.688
CSP($K=4$)+LDA (gated)	Motor imagery	CV acc 0.832 $\pm$ 0.072
CSP($K=4$)+SVM/RBF	Motor imagery	Eval acc=0.743
Ridge (scalar r , LOO)	Drift estimation	$R^2 = 0.997$
Multi-Ridge (α, γ)	Axis recovery	$R^2 = 0.997/0.990$
Multi-Ridge (β)	Noise floor	$R^2 = 0.443$
Multi-Ridge (δ_{ms})	Timing jitter	$R^2 = -0.214$
KMeans $k = 3$	Trajectory clustering	Sil.=0.535
KMeans $k = 2$	Domain profiling	Sil.=0.821

TABLE IV: FastAPI Serving Endpoints.

Endpoint	Method	Description
/predict/trial	POST	Gate + LDA on single epoch
/predict/upload	POST	Batch inference on .npz file
/calibrate/subject	POST	Adaptive calibration
/analyze/session	POST	Session ϕ_s and $\hat{r}$
/health	GET	Docker liveness check
/metrics	GET	Prometheus scrape endpoint

VII. CI/CD PIPELINE

The two-tier GitHub Actions architecture separates fast-feedback from thorough validation. The `ci.yml` workflow executes on every push and pull request: Ruff linting, mypy type checking, pytest against synthetic EEG fixtures (no MOABB download, <7 min end-to-end), a synthetic model smoke-test via `seed_artifacts.py`, and a Docker API image build without a registry push.

The `train.yml` workflow is triggered via `workflow_dispatch` or a weekly cron (Sunday 04:00 UTC). It caches the `~/mne_data` MOABB directory between runs, executes the full BNCI2014_001 training pipeline (15–45 min), runs DeepChecks pre-training data integrity and post-training model quality suites, uploads `artifacts/registry/` as a GitHub artifact, and pushes a versioned Docker image to the GitHub Container Registry (GHCR) tagged with both the commit SHA and `:latest`. A failing DeepChecks model suite blocks the registry write.

VIII. DISCUSSION

A. Gate Calibration and the Paradox Problem

The acceptance gate paradox was observed in 2 of 9 LOSO subjects (S8, S9), both near ceiling accuracy (≥ 0.924). A permissive threshold (`cal_threshold` 0.408 and 0.285, respectively) admits nearly all trials, so the few rejected trials fall in well-calibrated LDA regions. The paradox is absent at low accuracy, limiting its clinical concern. A subject-adaptive threshold, calibrated against held-out calibration trials rather than a fixed percentile rule, would reduce paradox prevalence. The non-monotonic relationship between `cal_threshold` and eval accuracy confirms that the gate proxies artifact-based signal integrity, not motor imagery discriminability.

B. Classifier Blindness to Drift

The near-zero permutation importance of classifier-derived features (`lda_session_acc`, `lda_conf_mean`) in both the scalar r regression and domain profiling constitutes a principal scientific contribution: LDA confidence can remain high while the underlying signal drifts, because the classifier adapts its decision boundary to the degraded distribution. Raw signal quality features anchored to absolute physiological plausibility (`p2p_max_mean`, `frob_dist_to_baseline`) are strictly more reliable drift indicators than classifier outputs (Fig. 5).

C. Timing Jitter and Cross-Prior Limitations

The δ_{ms} axis failure ($R^2 = -0.214$) demonstrates that timing jitter introduces perturbations orthogonal to session-level covariance statistics. Addressing this axis requires trial-level latency features or a dedicated temporal alignment model. The cross-prior generalisation gap of 0.398 indicates that drift models calibrated on one patient population’s degradation profile do not transfer reliably to populations with different degradation characteristics, motivating a Bayesian or prior-agnostic estimation approach.

IX. CONCLUSION AND FUTURE WORK

NeuroDrift demonstrates that a three-stage gated pipeline materially improves on standard single-session BCI calibration. The frozen acceptance gate produces a +4.2 pp accuracy gain over the unfiltered baseline, confirmed by the accepted vs. rejected accuracy differential (0.882 vs. 0.800). LOSO evaluation reveals $\mu = 0.728$ ($\sigma = 0.174$), quantifying the inter-subject heterogeneity that single-subject metrics conceal. Ridge LOO regression achieves $R^2 = 0.997$ (RMSE=0.022), and unsupervised domain profiling achieves silhouette=0.821 without session-level labels. The finding that classifier accuracy is a poor proxy for signal quality ($d = 1.011$, $p = 0.238$; $\Delta R^2 \approx 0$ in drift regression) is a principal scientific contribution.

Immediate research priorities include: (i) replacing CSP+LDA with Riemannian minimum-distance-to-mean (MDM) classification [5]; (ii) implementing subject-adaptive gate thresholding via a resting-state quality probe; (iii) enriching the session feature vector with trial-level latency features to recover the δ_{ms} axis; and (iv) deploying a Gaussian

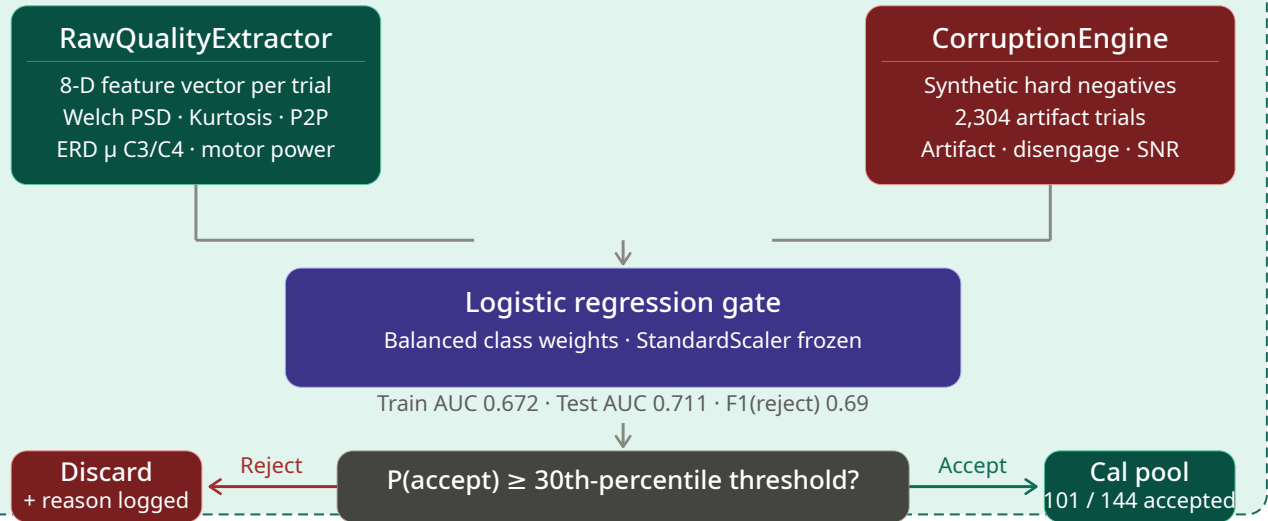
Process regressor for uncertainty-aware r estimation. Longer-term milestones include federated longitudinal drift adaptation across multi-site deployments and clinical validation against Fugl-Meyer recovery scores in stroke rehabilitation cohorts.

■ ML model ■ Data / signal ■ Artifacts ■ Degradation ■ Decision ■ Infrastructure

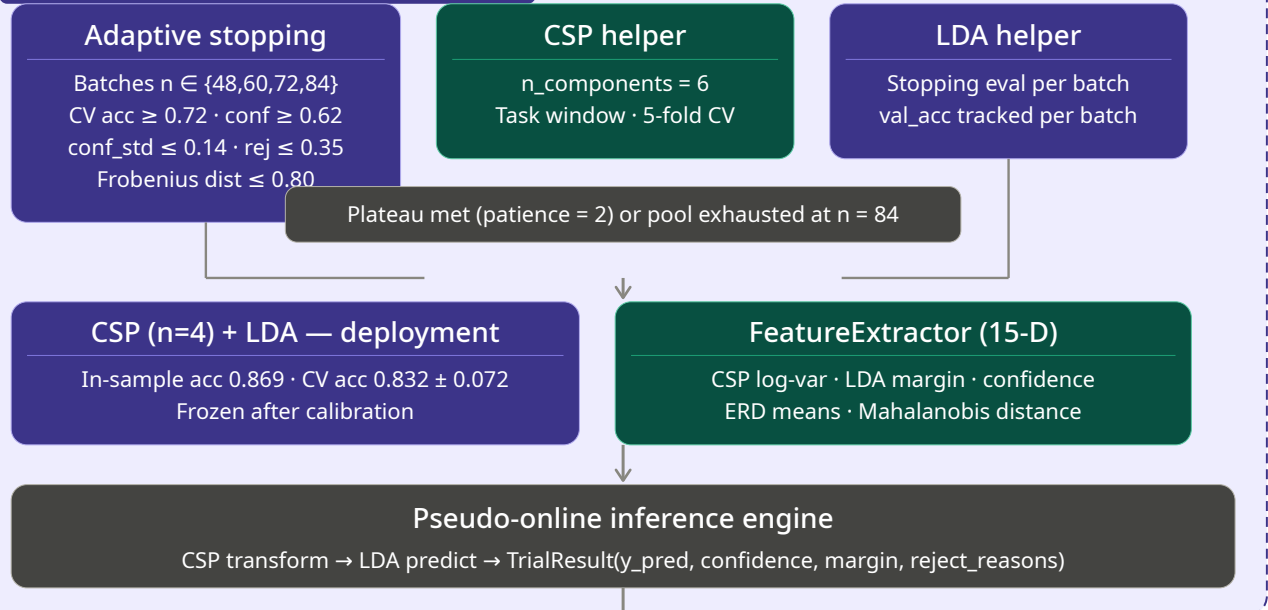
Data ingestion



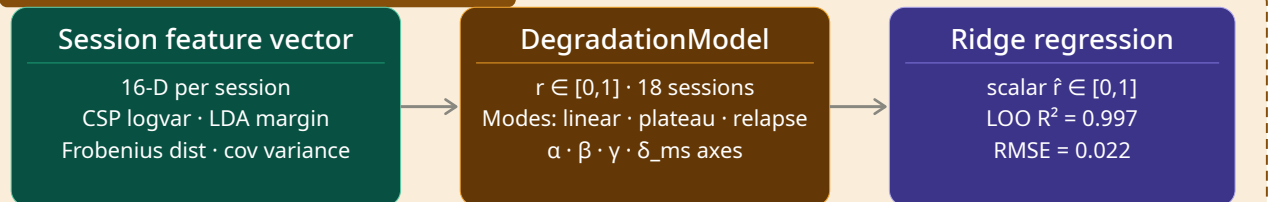
Stage 1 — Global acceptance gate



Stage 2 — Adaptive per-subject calibration



Stage 3 — Longitudinal drift tracking



REFERENCES

- [1] S. Soekadar et al., “EEG/EOG-based hybrid brain neural computer interaction during daily routine,” *Front. Neurosci.*, vol. 9, 2015.
- [2] V. Jayaram and A. Barachant, “MOABB: Trustworthy algorithm benchmarking for BCIs,” *J. Neural Eng.*, vol. 15, no. 6, p. 066011, 2018.
- [3] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, “Optimal spatial filtering of single trial EEG during imagined hand movement,” *IEEE Trans. Rehab. Eng.*, vol. 8, no. 4, pp. 441–446, 2000.
- [4] F. Lotte et al., “A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update,” *J. Neural Eng.*, vol. 15, no. 3, p. 031005, 2018.
- [5] M. Congedo, A. Barachant, and R. Bhatia, “Riemannian geometry for EEG-based brain-computer interfaces: A primer and a review,” *Brain-Computer Interfaces*, vol. 4, no. 3, pp. 155–174, 2017.
- [6] V. Lawhern et al., “EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces,” *J. Neural Eng.*, vol. 15, no. 5, p. 056013, 2018.
- [7] B. Blankertz et al., “The BCI Competition 2003: Progress and perspectives in detection and discrimination of EEG single trials,” *IEEE Trans. Biomed. Eng.*, vol. 51, no. 6, pp. 1044–1051, 2004.
- [8] M. Atzori et al., “DeepChecks: A library for testing and validating machine learning models and data,” *arXiv preprint arXiv:2203.00888*, 2022.